

Carrum Health Data Engineering Challenge

Please spend no longer than 2 hours on the challenge. Feel free to make assumptions where necessary. There is no right answer – we're interested in seeing your systems design skills, familiarity with different technologies, and approach to ETL and problem solving.

Introduction & Requirements

We'd like you to design a system to empower data-driven marketing campaigns utilizing data from several sources.

The marketing team is trying to help maximize button clicks in a very simple (pretend) app, where a user can only do three things: **click on button A**, **click on button B**, and **click on button C**.

Imagine we have the following app:



Marketing thinks that people who have already clicked on a button will be more likely to click on it in the future – e.g. people who clicked on A will click on A again. To that end, they are trying to market button A to folks who have already clicked on button A, button B to button B clickers, etc.

As a data engineer, we need to deliver the following requirements to the marketing platform:

- 1. Recent (no older than 24 hours) click data with the following schema:**
 - Data is stored in the platform in “attributes” per member profile. It's able to accept any number of attributes defined in the following schema: `{user: username (string), attributes: {attribute1: (any type), attribute2: (any type), ...}}`. Attributes will be overwritten or added automatically each time a new payload is received for a member.
 - Marketing wants to utilize the **number of clicks on A, B and C for each member in the last week** for their upcoming campaign.
 - Data should always be at least 24 hours fresh.
 - Users only need to be considered who have ever clicked on A, B, or C.
- 2. The above schema can be ingested into the marketing platform in two ways:**
 - Bulk CSV upload via SFTP (schema reflects the above data `{user, attribute1, attribute2, ...}`)

- HTTP POST request for each member with a JSON payload mirroring the above schema. This will synchronously update the user record in the marketing platform using your `attribute` variables.

Data Description

The data originates from the following three sources:

- **Source 1: 1st party app backend (Postgres DB)**
 - Schema:
 - Clicks table: {`userName: string, clickTarget: enum(A, B or C), clickTime: timestamp`}
 - The database has a primary instance & read-only replicas
- **Source 2: Realtime API for 3rd party most-recent-click tracker for A or B**
 - Users could have clicked on A or B (but not C) in another app. Marketing has validated that these external A and B clicks are as valid as 1st party clicks.
 - HTTP GET requests are supported, with JSON records.
 - Records are queried with `userName` (same names between 3rd party and your 1st party app) and contain the most recent click time for A and B:
 - {`"userName": string, "recentClickA": timestamp, "recentClickB": timestamp`}
 - 3rd party API records are updated synchronously when a user clicks on A or B within their app.
- **Source 3: Batch CSV dumps for 3rd party historical clicks on C**
 - Users could have clicked on C in another app. Marketing has validated that external C clicks are as valid as 1st party C clicks.
 - CSV files contain all historical clicks (partitioned into multiple files as needed) for users on their platform.
 - The schema is simple: {`userName: string, clickCTime: timestamp`}
 - The files are dumped in a cloud bucket on a regular basis, once per day, but not always at the same time.

Data & Design Considerations

- For scale, assume there are roughly ~10 million users to deliver data
- Try to optimize for solutions that will cost <\$5k per month

Challenge Assignment

Using data from all three sources, design a system to deliver data to the marketing platform which meets requirements 1 and 2 above. You can leverage any existing cloud technologies or tech stacks that you think would be appropriate.

Please include the following:

- a system diagram
- a short explanation of design choices and technologies in the diagram
- descriptions of the processes and schemas that you would use for the data at each step (we'll want to be able to see and understand your schema(s))
- deployment / productionalization strategy for monitoring and any necessary updates to the pipeline
- consideration of inherent problems with data quality or reliability from any of the sources or pipeline requirements
- drawbacks of your solution

We'll also want to discuss

- whether this is an ETL or an ELT solution. How would you switch from one to the other? What are the benefits and disadvantages of each?
- whether this is a from-scratch solution or leverages more out-of-the-box technologies. How would you switch from one to the other? What are the pros and cons of each?

Feel free to simply make a Google document or slide deck which walks through the above bullets. You can use any system's diagram tool you're comfortable with. (Free options include excalidraw or Google Slides diagrams.)

After completing, the next step will be to walk through the document(s) with the Carrum data engineering team where you'll have 20 minutes to present, followed by 20 minutes of questions and discussion.